

Datasets

1. Dataset Name: IaC-Eval

- Source: Georgia Tech (NeurIPS 2024 Datasets & Benchmarks Track), hosted on Hugging Face / GitHub
- URL: <https://github.com/autoiac-project/iac-eval>
- Size: 458 human-curated Infrastructure-as-Code questions, ranging from simple to complex [GitHub](#)
- Number of Records: 458 scenarios
- Number of Features: Each record includes a natural-language problem description, an infrastructure-intent specification, and reference correctness checks (a natural language IaC problem description and an infrastructure intent specification used to verify if the generated program conforms to the user's intent)
- Data Type: Text (natural language) + Terraform HCL code
- License: Dataset released under Creative Commons Attribution 4.0; accompanying code under MIT license
- Purpose of Using the Dataset: Benchmark the IaC-generation and syntax-validation agent's ability to produce/validate correct, deployable Terraform code against known-correct infrastructure intent
- Preprocessing Required: Parse HCL into structured resource graphs; align each scenario's intent specification with your framework's internal validation schema; filter by target AWS services relevant to smart-manufacturing use cases

2. Dataset Name: Multi-IaC-Bench

- Source: arXiv preprint (2025), synthetic generation pipeline over public GitHub templates
- URL: <https://arxiv.org/abs/2509.05303>
- Size: Triplet-structured dataset spanning three IaC formats
- Number of Records: Multiple template–request–updated-template triplets consisting of initial IaC templates, natural language modification requests, and corresponding updated templates, created through a synthetic data generation pipeline with rigorous validation [arXiv](#)
- Number of Features: 3 core fields per record (original template, NL modification request, resulting template) plus metadata (format type, provider)

- Data Type: Text (CloudFormation, Terraform, and CDK code) + natural language
- License: Open academic release (check repository for specific terms at time of use)
- Purpose of Using the Dataset: Directly supports your framework's multi-format requirement — evaluate how well the agents parse and validate CloudFormation, Terraform, and CDK consistently, since the dataset was built specifically to address the lack of benchmarks across multiple IaC formats [arXiv](#)
- Preprocessing Required: Normalize the three formats into a common intermediate representation for cross-format agent evaluation; split by format for per-technology accuracy reporting

3. Dataset Name: TerraGoat (Vulnerable-by-Design Terraform)

- Source: Bridgecrew (Palo Alto Networks), open-source GitHub repository
- URL: <https://github.com/bridgecrewio/terragoat>
- Size: Full multi-cloud Terraform project (AWS, GCP, Azure modules)
- Number of Records: Dozens of intentionally misconfigured resource blocks across services like EC2, S3, IAM, RDS, EKS
- Number of Features: Resource type, misconfiguration category, associated policy/rule ID (when cross-referenced with a scanner like Checkov)
- Data Type: Terraform HCL code (intentionally insecure configurations)
- License: Open-source (check repo LICENSE file; Bridgecrew tooling is Apache-2.0 style, commonly used for benchmarking without restriction)
- Purpose of Using the Dataset: Built to enable DevSecOps teams to design and test misconfiguration-prevention strategies — use it as ground truth to test your security-assessment agent's detection rate on IAM, network, and storage misconfigurations, since it is less about runtime exploitation and more about learning how insecure Terraform looks in review, scanning, and policy enforcement [GitHubProduct-security](#)
- Preprocessing Required: Tag each resource block with its known vulnerability type (cross-reference with Checkov/tfsec rule outputs) to create ground-truth labels for precision/recall evaluation

4. Dataset Name: Runtime Telemetry & Drift Dataset (project-generated)

- Source: Self-collected from Amazon CloudWatch, AWS Config, and deployment logs within your own AWS test environment
- URL: Not applicable — generated internally during project implementation

- Size: Depends on test deployment duration/scale (recommend at least several weeks of continuous logging across multiple stacks)
- Number of Records: One record per resource-state snapshot or drift event
- Number of Features: Resource ID, expected state (from IaC template), observed state (from CloudWatch/Config), timestamp, drift type, cost delta
- Data Type: Structured JSON/time-series logs
- License: Internal/project-owned (no external licensing, but note AWS service terms apply)
- Purpose of Using the Dataset: This is the dataset that makes your "telemetry-guided reasoning" and configuration-drift-detection objectives testable — none of the public benchmarks above include real runtime telemetry, so this fills that specific gap in your framework
- Preprocessing Required: Time-align telemetry with corresponding IaC deployment versions; normalize AWS Config drift events into a schema matching your template definitions; handle missing/incomplete CloudWatch metrics